We Let Six AI Agents Prep a CRAN Package

orthoMTL — and the skill files that changed nothing

Kevin Vervier
Novartis · R/Pharma 2026

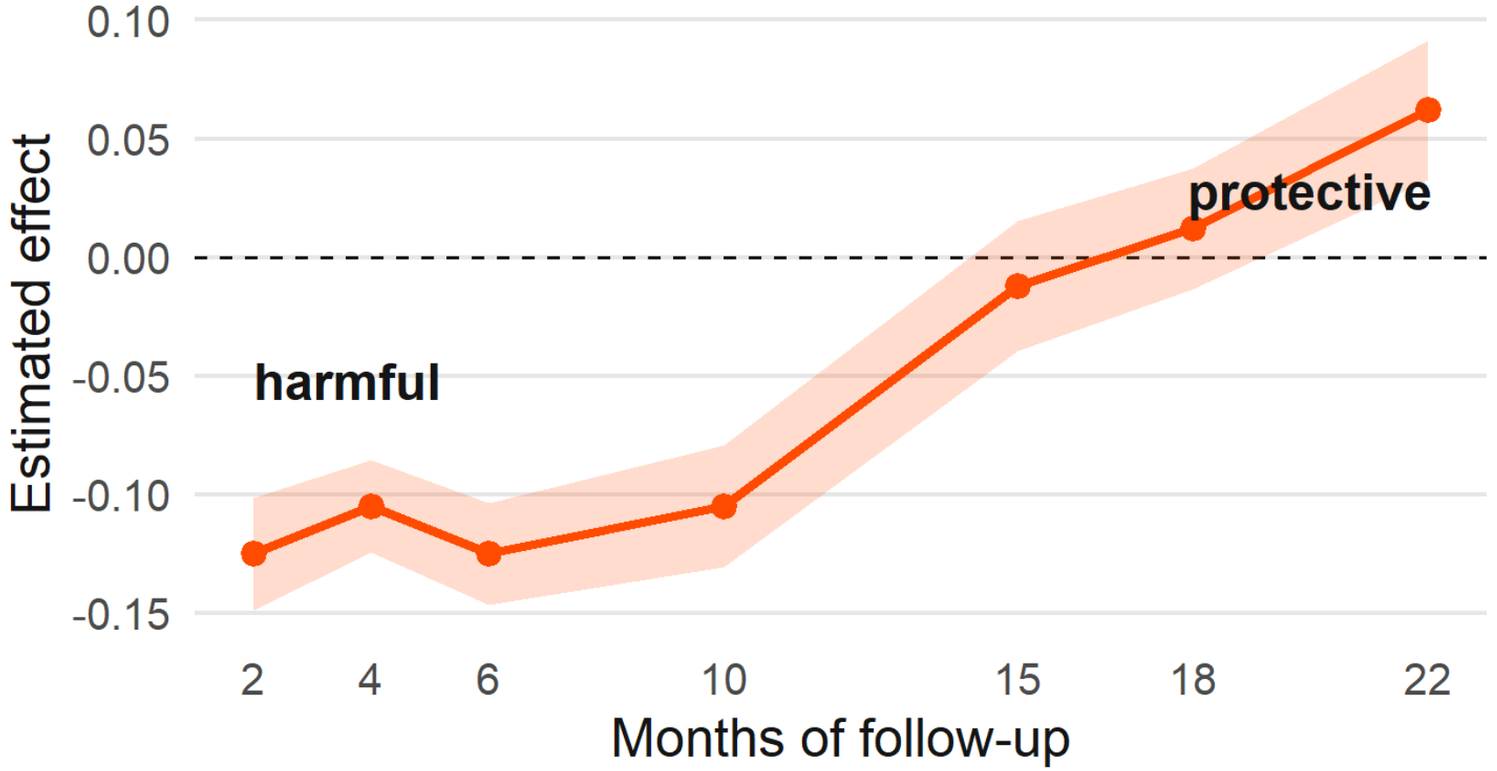
October 20, 2026

NOVARTIS

A biomarker that changes its mind

In a late-stage oncology trial, one baseline biomarker looked like **nothing**.

A standard survival model scored its effect at ≈ 0.



Schematic of one biomarker's effect. Shape is illustrative.

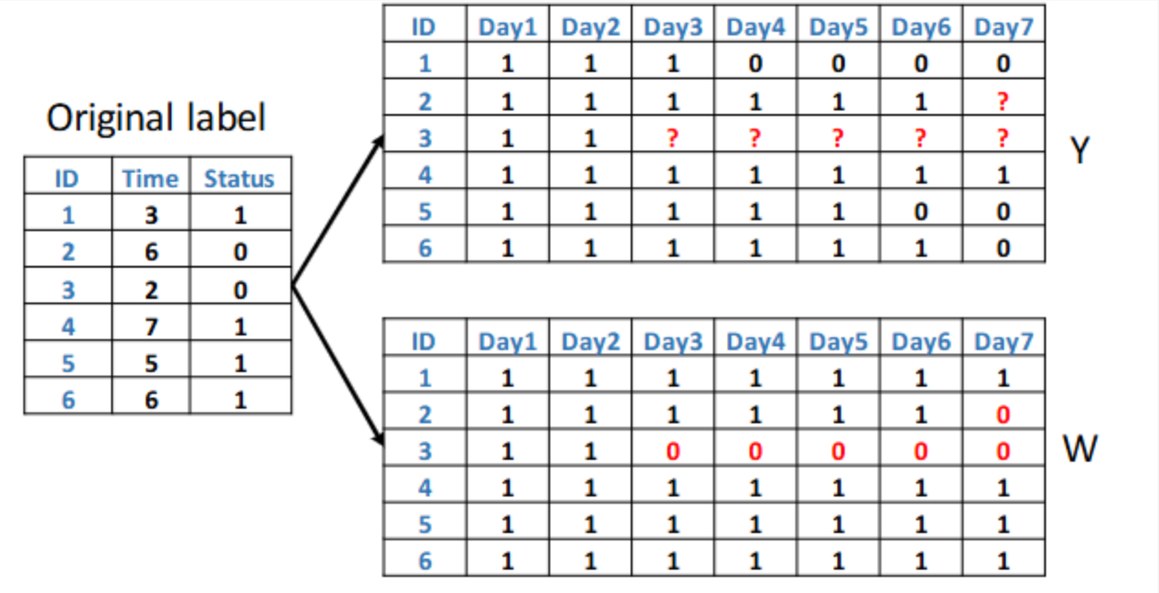
Harmful early. Protective late. The two halves cancelled out.

NOVARTIS

orthoMTL: don't average over time

Cut follow-up into **landmarks**. Fit all of them **at once**, letting each landmark keep its own effect.

- Survival → a series of yes/no tasks
- One model, not seven
- Oncology application: C-index **0.66 vs 0.63**



NOVARTIS

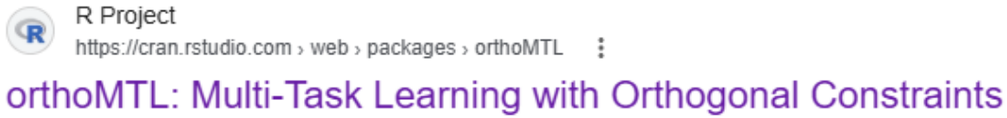
`github.com/Novartis/orthoMTL` — on CRAN, 2026

From PhD code to CRAN

Twelve-year-old research code. An agent did the packaging work.

It also found a **real bug** in my own 2014 gradient — labels were ± 1, the code used the $0/1$ formula.

Caught because flipping every label **must** flip the answer. It didn't.



Then CRAN rejected us. Twice.

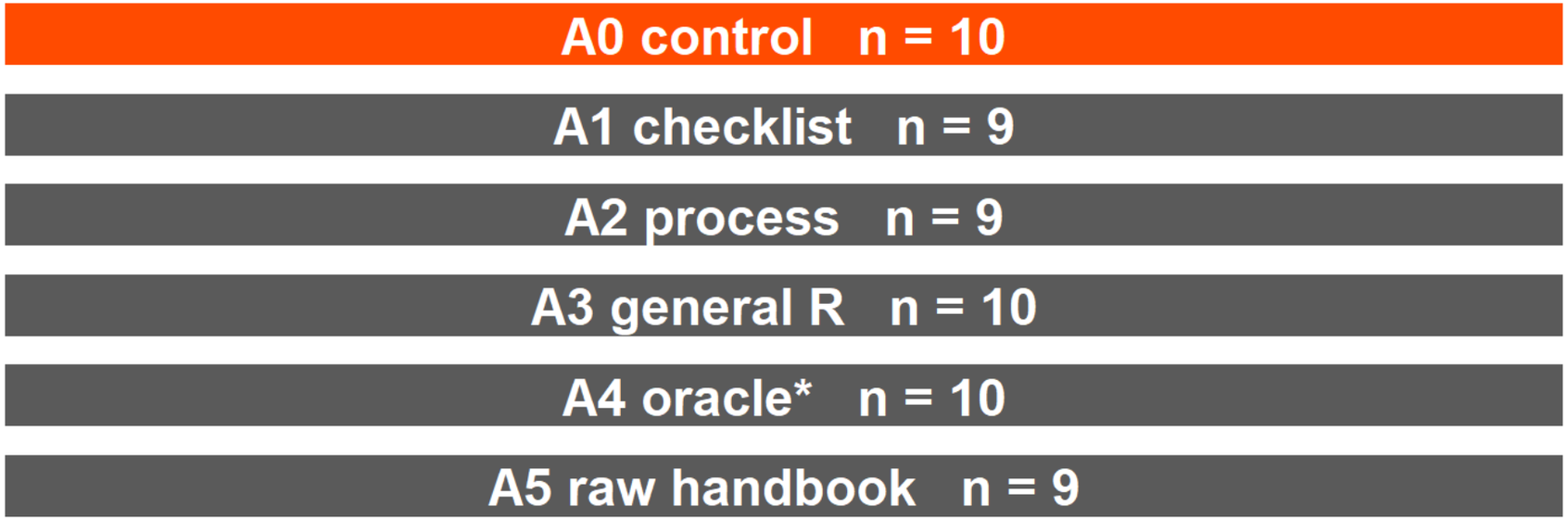
1. An invalid `LICENSE` field.
2. A hard-coded `set.seed()`.

Neither was caught by any automated check. Both were caught by **a human**.

So: would a “skill file” have saved me?

Skill files are everywhere. **Does anyone measure them?**

Re-break the package with four classic CRAN defects. Hand it to an agent under **six conditions**. Frozen prompt, fresh copy each time, results **blinded** before scoring.



NOVARTIS

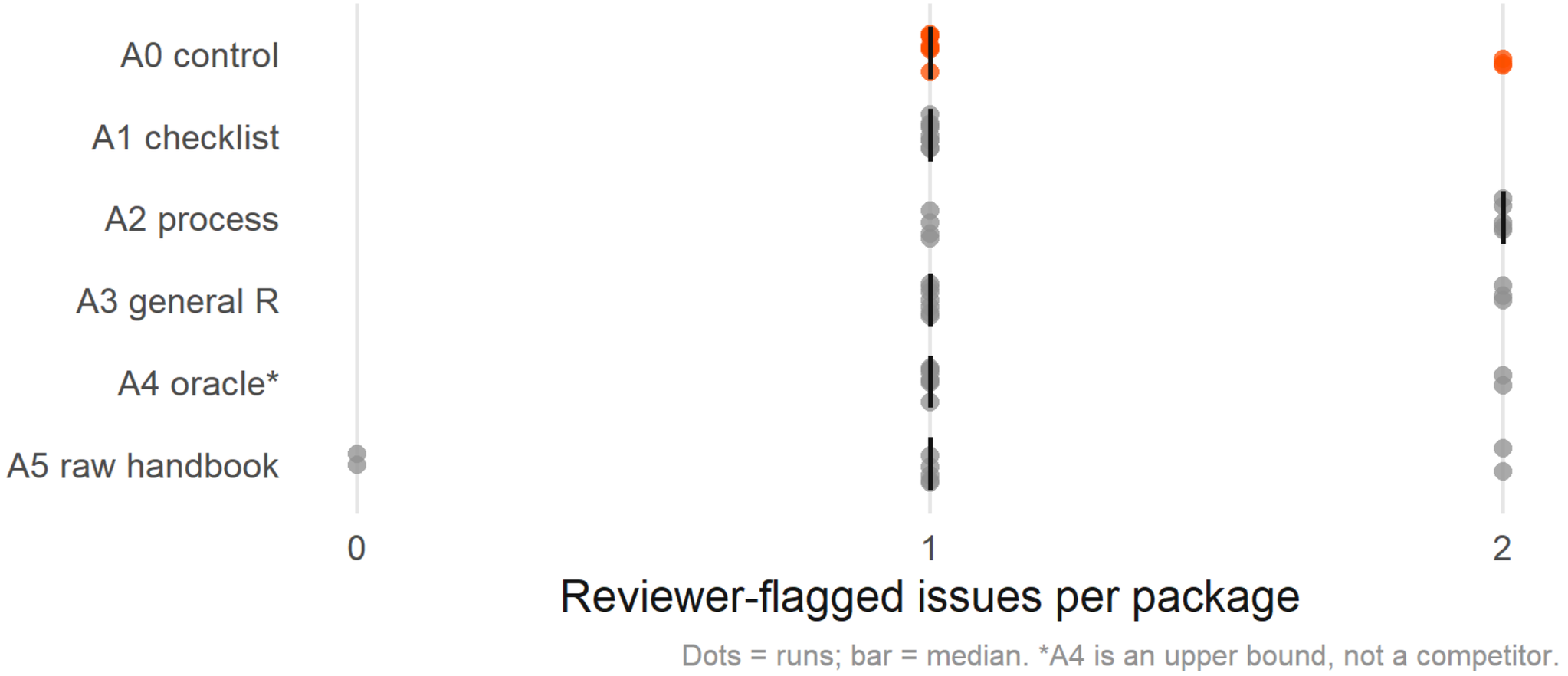
The result

0

difference between the best skill file and **no skill file at all**

All **57/57** runs fixed all four defects — including the control. Even the arm I wrote from our own failures.

Every arm lands in the same place



Orange = **no skill file**. It is not behind anyone.

NOVARTIS

Why the control did so well

The agent runs `R CMD check` and reads its own failures.

A machine oracle beats any checklist — it **can’t be out of date**.

Skill files add nothing where a tool already tells the truth.

What I am *not* claiming

- Every arm could read the CRAN docs anyway → “**no benefit beyond docs at hand**”, not “skills are useless”
- **One** package, **one** agent, **one** model version
- I extended the sample size **after** peeking — post-hoc, not pre-registered
- No positive control, so I can’t prove the measurement *could* detect a win
- 3 runs lost to rate limits (57 of 60 analysed)

Take home

1. If effects can change over time, don’t average them away → `orthoMTL`

2. Agents are good at checks a **machine** can run. Buy those, not checklists.

3. The hard bugs needed **me** to know which symmetry had to hold.

Generation got cheap. Verification did not.

`github.com/Novartis/orthoMTL` · Vervier et al. 2014 · Juric et al. 2026

Backup

NOVARTIS

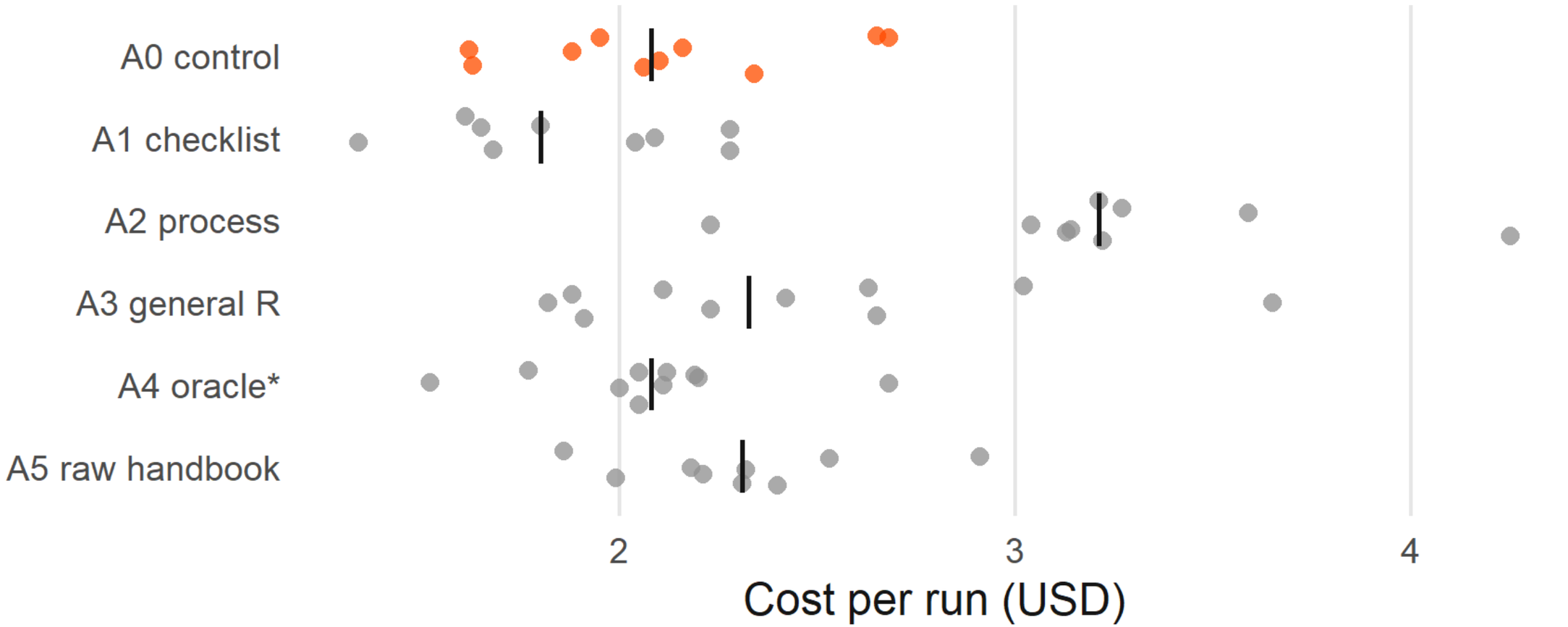
Backup: what the six arms were

Arm	Skill file
A0	none (control)
A1	CRAN checklist
A2	CRAN process + adversarial review
A3	general R-package development
A4	written from *this* package's failures
A5	raw handbook text

Backup: the three ways we checked correctness

Check	What it proves
Against the old validated package	identical coefficients on 5 fixed cases
Flip every label	answer must flip; `0.62` → `0.000`
Simulated data, known truth	recovers the right features, 10/10 seeds

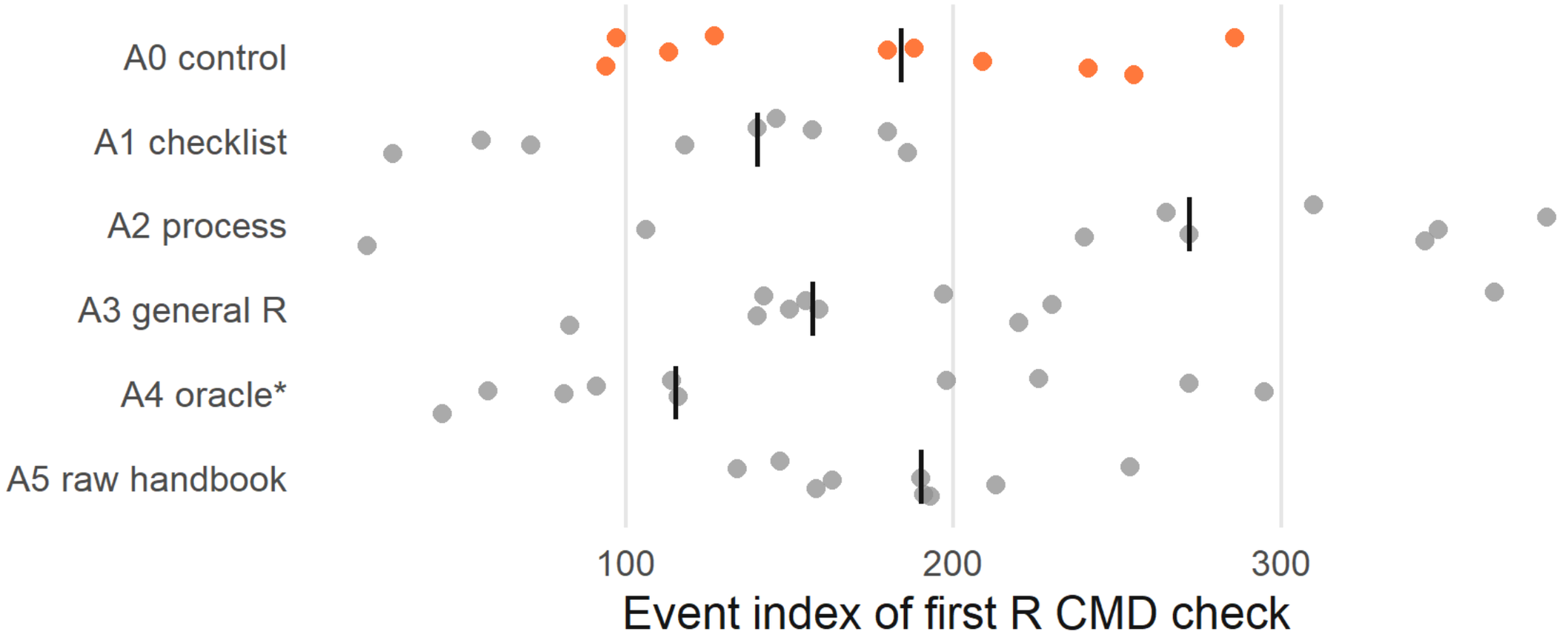
Backup: cost per run



Dots = runs; bar = median. *A4 is an upper bound, not a competitor.

NOVARTIS

Backup: when did the agent first run `R CMD check`?



Dots = runs; bar = median. *A4 is an upper bound, not a competitor.

NOVARTIS

Backup: what I would change

- Seed defects only a **human reviewer** could catch
- Add a deliberately handicapped arm as a positive control
- Judge each package several times, not once
- More packages, more models